AS · Cambridge (CIE) · Physics

2 hours 7 questions

Structured Questions

Non-Uniform Motion

Drag Force & Air Resistance / Terminal Velocity

Easy (2 questions)	/23
Medium (3 questions)	/47
Hard (2 questions)	/21
Total Marks	**/91**

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Easy Questions

1 (a) A skydiver of mass 75 kg jumps from an aircraft.

(i) State the difference between the mass of a body and its weight.

[2]

(ii) Calculate the weight of the skydiver.

[2]

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..... **(4 marks)**

(b) The graph in Fig. 1.1 shows the vertical velocity of the skydiver during the first 40 s of the fall

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before opening his parachute.

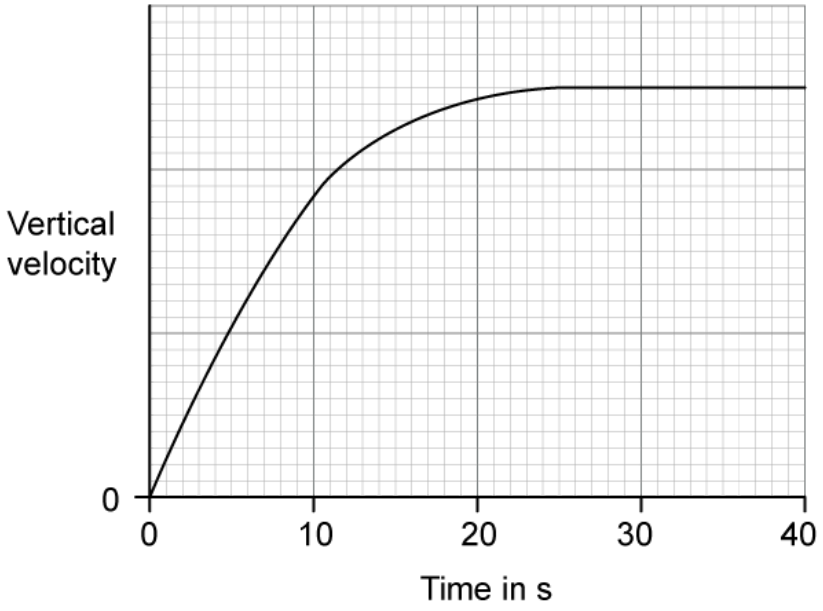


Fig. 1.1

Explain the shape of the graph

(i) between 0 and 5 s

[2]

(ii) between 5 and 25 s

[2]

(iii) between 25 and 40 s

[2]

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(6 marks)

(c) Fig. 1.2 shows the skydiver falling at a constant velocity.

Add labelled arrows to Fig. 1.2 to show the forces acting on the skydiver.



Fig. 1.2

(3 marks)

(d) The skydiver opens his parachute after 40 s.

On Fig. 1.3 below, complete the graph to show how the skydiver's vertical velocity changes after 40 s.

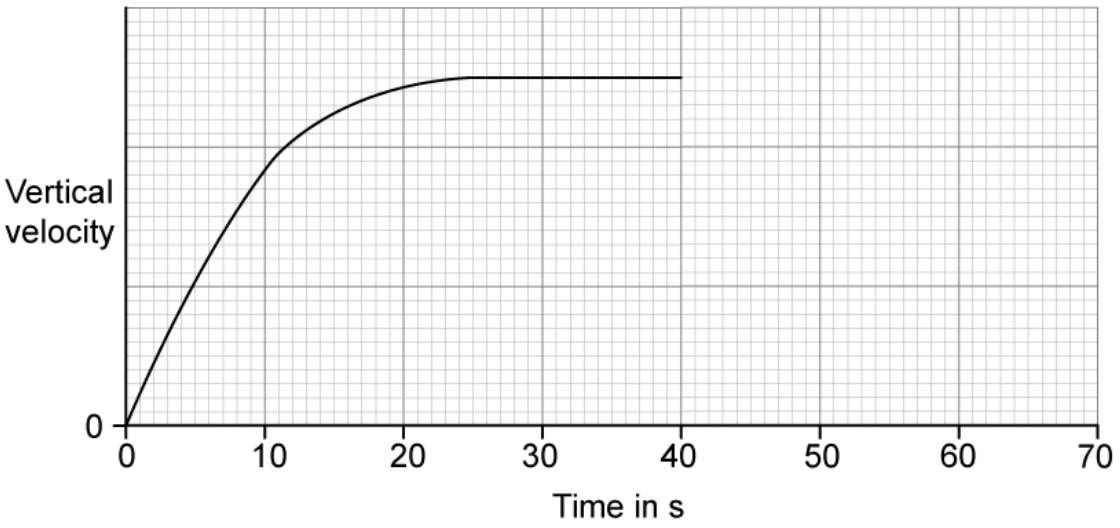


Fig. 1.3

(2 marks)

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2 (a) In physics, the motion of an object falling through the atmosphere is often studied. Fig. 1.1 shows a diagram of this motion.

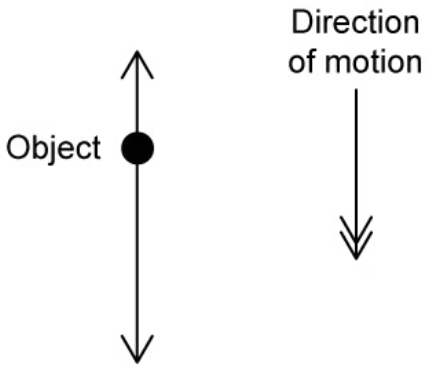


Fig. 1.1

The object shown moves vertically downwards through the atmosphere.

Identify the two forces acting on the object and label them on the diagram.

(2 marks)

(b) Often, a simplifying condition is assumed so that, in these cases, only a single force acts on objects as they move through the atmosphere.

State the simplifying condition that is normally assumed.

.....

(1 mark)

(c) Terminal velocity is only attained if both forces act on the object.

State and explain the magnitude of the resultant force on the object if it moves at its terminal velocity.

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(2 marks)

(d) Sketch a graph on the axes in Fig. 1.2 to show an object that is released from rest at $t = 0$ s

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and falls vertically through the atmosphere, attaining terminal velocity, $v_{terminal}$ after t = 4 s.

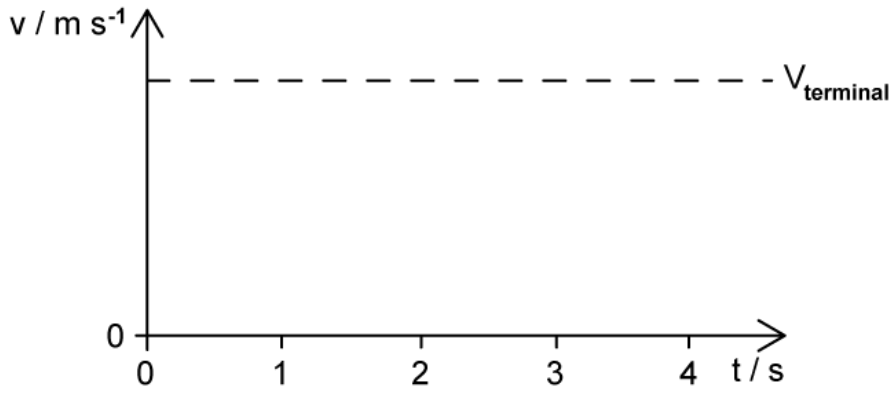


Fig. 1.2

The line corresponding to $v_{terminal}$ is included as guidance.

(3 marks)

Medium Questions

1 (a) Fig. 1.1 shows a stone block being pulled up a slope at a constant speed by a cable attached to an electric motor.

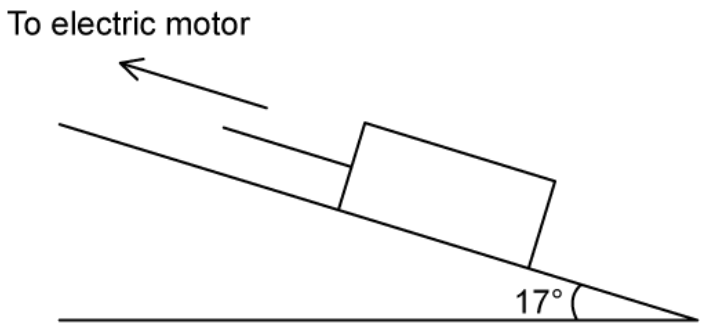


Fig. 1.1

The slope is inclined at an angle of 17° with the horizontal. The mass of the block is 250 kg, and the tension *T* in the cable is 1.2 kN.

On Fig. 1.1, draw and label the forces acting on the block.

(4 marks)

(b) With reference to the motion of the block, discuss whether the block is in equilibrium.

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..... **(2 marks)**

(c) The total resistive force *R* acting on the box is constant throughout its motion on the slope.

Calculate the magnitude of the resistive force *R*.

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..... **(4 marks)**

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(d) The cable connecting the block to the electric motor abruptly breaks.

Calculate the acceleration of the block as it moves down the slope.

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..... **(3 marks)**

2 (a) A student uses a motion sensor connected to a laptop to investigate the motion of a hollow ball of mass 1.5×10^{-2} kg falling through the air.

The ball is dropped from rest. It reaches terminal velocity before it reaches the ground.

The upthrust on the ball is negligible.

Explain briefly why the acceleration of the ball

(i) decreases with time

[2]

(ii) is 9.8 m s^{-2} initially.

[1]

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..... **(3 marks)**

(b) The variation with time *t* of the velocity *v* of the ball as it falls towards the ground is shown in

Fig. 1.1.

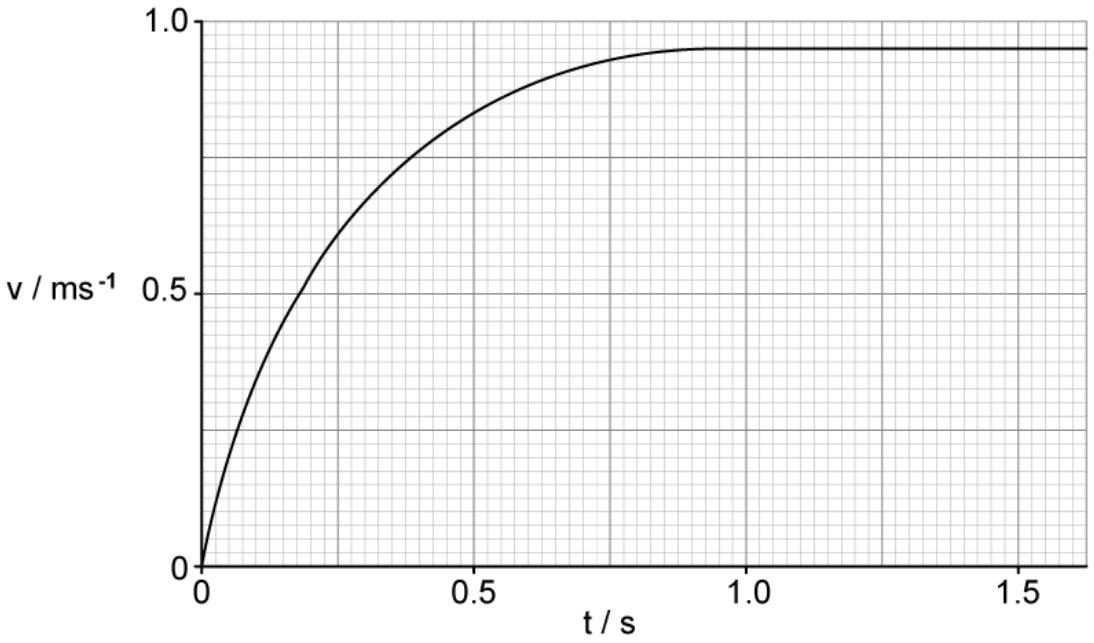


Fig. 1.1

(i) Use Fig. 1.1 to determine the magnitude of the acceleration of the ball at time t = 0.25 s.

[3]

(ii) Calculate the resultant force acting on the ball at time t = 0.25 s.

[2]

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..... **(5 marks)**

(c) Use your answers in **(b)** to determine the drag on the ball at time t = 0.25 s.

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(3 marks)

(d) The student adds a small amount of sand inside the hollow ball and repeats the experiment. The ball is dropped from rest and reaches terminal velocity before it reaches the ground.

(i) Describe how the forces acting on the ball filled with sand at $v = 0.50$ m s^{-1} compare with the forces acting on the hollow ball at this speed.

[2]

(ii) State and explain how the terminal velocity of the ball filled with sand will differ from the terminal velocity of the hollow ball.

[3]

(5 marks)

3 (a) On an alien planet similar to Earth, a parachutist jumps out of an aircraft.

The graph in Fig. 1.1 shows how the vertical speed of the parachutist changes with time during the first 20 s of his jump. To avoid air turbulence caused by the aircraft, he waits a short time after jumping before pulling the cord to release his parachute.

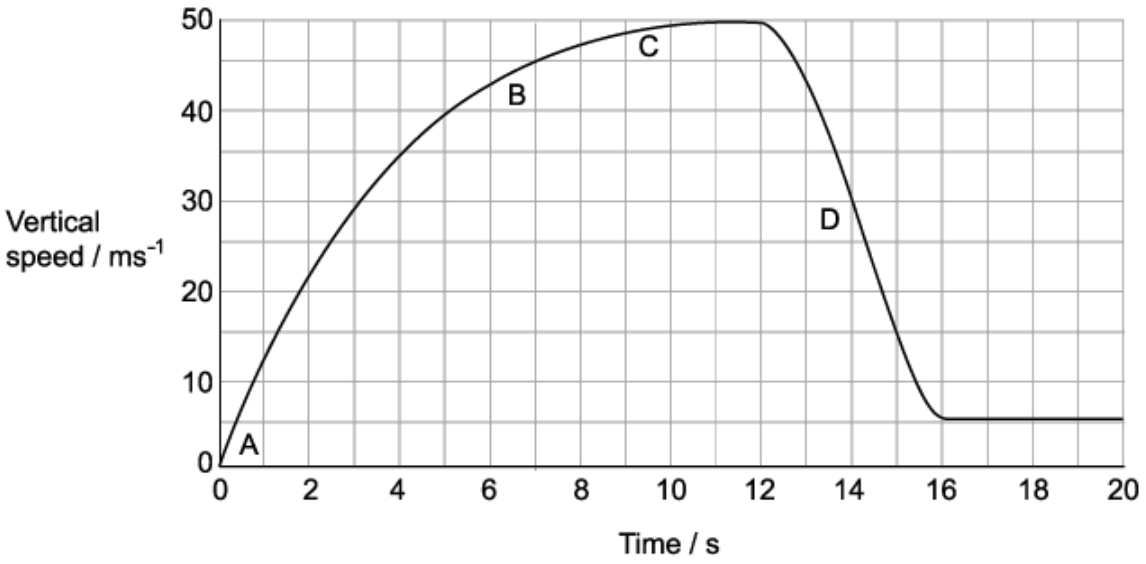


Fig. 1.1

The parachute opens after 12 s.

With reference to the forces acting on the parachutist, explain the shape of:

(i) Region **A**

[3]

(ii) Region **B**

[3]

(iii) Region **C**

[3]

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(9 marks)

(b) Determine the acceleration of free fall on this planet.

(1 mark)

(c) On Fig 1.1, draw graph lines for the following scenarios:

(i) Scenario 1: air resistance is zero.

[2]

(ii) Scenario 2: the parachute deployed is half the size of the original scenario. You may assume that air resistance is directly proportional to the surface area of the parachute.

[2]

(4 marks)

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(d) A light breeze exerts a constant horizontal force on the parachutist. He eventually reaches a constant horizontal speed.

Explain why this horizontal terminal speed he reaches is much lower than the vertical terminal speed, before the parachute is opened.

(4 marks)

Hard Questions

1 (a) Strongman competitions involve competitors performing various challenges where they have to produce very high forces.

In one particular event, a competitor must drag a cement block across a rough floor, using a rope.

When an object is moving, the frictional force is proportional to the normal reaction force of the object.

In the first round of the competition, the competitor must drag a 200 kg block across the rough surface. In the second round, they must drag a new 280 kg block across the same surface. The force exerted by the competitor in round 2 is 1650 N.

Calculate the difference in the frictional forces between the first and second rounds, when the blocks are in motion.

You may assume the block moves with constant speed and air resistance can be ignored.

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..... **(4 marks)**

(b) Justify the decision to ignore air resistance in part (a).

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..... **(3 marks)**

(c) Recall from part (a) that the competitor exerts a horizontal force of 1650 N on the 280 kg block.

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They pull the rope, however, at a 15° angle to the horizontal plane, as shown in Fig. 1.1.

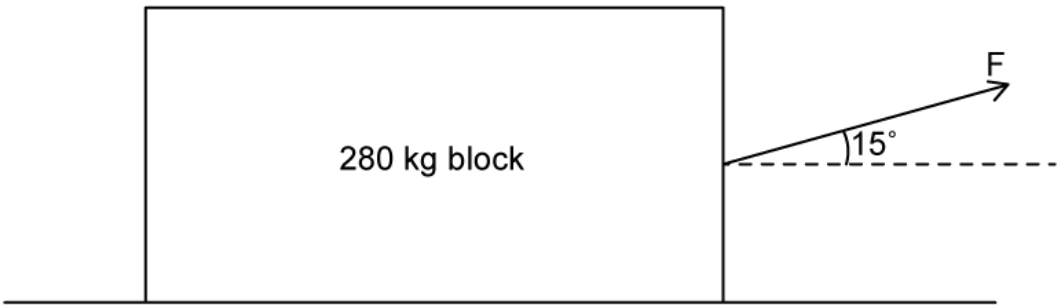


Fig. 1.1

Calculate the vertical force the competitor exerts on the block.

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..... **(2 marks)**

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2 (a) Air resistance can be modelled using the following equation:

$$F_v = kAv^2$$

where F_V is the force due to air resistance at a given velocity v, A is the surface area of the object experiencing the air resistance and k is a constant.

Two balls are released from a high bridge. Ball A has a volume 27 times larger than ball B.

Compare the forces of air resistance on each ball when they reach a speed of 20 m s^{-1}.

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..... **(3 marks)**

(b) The experiment is repeated, but this time with two balls of identical size. One, however, has a greater mass.

A passer-by claims that the mass difference will not affect the balls landing at the same time.

(i) Using Newton's second law and the equation from part (a), explain the relationship between acceleration and mass when air resistance is not ignored.

[3]

(ii) Explain why the passer-by is wrong.

[1]

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..... **(4 marks)**

(c) Determine the units of the constant k, in terms of SI units.

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(3 marks)

(d) Write an expression for terminal velocity, in terms of area A , mass m , acceleration of free fall g and the constant k.

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(2 marks)

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